A Frequency Decoupling Network for Semantic Segmentation of Remote Sensing Images

Xin Li, Feng Xu, Anzhu Yu, Xin Lyu, Hongmin Gao, *Senior Member, IEEE*, and Jun Zhou, *Senior Member, IEEE*

***Abstract*— Semantic segmentation of remote sensing images (RSIs) is vital for numerous geospatial applications, including land-use mapping, urban planning, and environmental monitoring. Traditional neural networks for semantic segmentation primarily focus on learning in the spatial domain, which often results in suboptimal performance due to the complexity of RSIs that exhibit diverse and intricate structures. To address this problem, we propose a novel frequency decoupling network (FDNet) that enhances feature representation by independently refining high-frequency and low-frequency components in the frequency domain. FDNet introduces three core components: a sparse-aware spectral enhancement module (SSEM) that optimizes spectral feature learning by compressing redundant information while highlighting informative spectral bands, a frequency decoupling attention module (FDAM) that precisely distinguishes and enhances high-frequency and low-frequency features and an attentive frequency context module (AFCM) that integrates SSEM and FDAM into a cohesive framework for enriched spectral context modeling. Extensive experiments conducted on four benchmark datasets demonstrate that FDNet outperforms several state-of-the-art methods, achieving superior segmentation accuracy and robustness across various terrains and imaging conditions. Ablation experiments further confirm the impacts of SSEM, FDAM, and AFCM.**

***Index Terms*— Attention mechanism, frequency domain, remote sensing images (RSIs), semantic segmentation.**

I. Introduction

SEMANTIC segmentation partitions an image into regions of the same class, labeling each pixel accordingly [1], [2], [3], [4], [5], [6]. Unlike image classification, which categorizes an entire image, semantic segmentation provides detailed delineations of different classes. Originating from image segmentation, it emphasizes semantic context, making segmented areas meaningful. This process is vital for many computer vision tasks and is used in various applications, including natural and medical image parsing [7], [8], [9], [10], autonomous driving [11], [12], [13], [14], road extraction [15], [16], [17], [18], and land-use mapping [19], [20], [21], [22], [23].

As imaging resolution increases, the complexity of information rises. Traditional machine learning algorithms, such as support vector machines (SVMs) and conditional random fields (CRFs), struggle with high-resolution remote sensing images (RSIs) [24], [25], [26]. The rapid development of deep learning has led to the application of convolutional neural network (CNN)-based methods for image segmentation [27], [28]. Fully convolutional networks (FCNs) introduced CNNs into image segmentation, achieving satisfactory results [29]. However, FCNs neglected multiscale spatial information, resulting in insufficient details [30]. U-Net [31] addressed this with skip connections, improving segmentation accuracy. Despite this, U-Net's skip connections struggle to fuse low and high-level features effectively [32]. Recognizing the importance of multiscale features, researchers developed DeepLab networks [33] to improve accuracy by expanding the receptive field without increasing parameters, using atrous spatial pyramid pooling (ASPP). Similarly, PSPNet [34] uses adaptive global pooling to enhance segmentation accuracy.

The integration of attention mechanisms [35], [36], [37] into CNNs has significantly advanced the field of semantic segmentation in RSIs [38], [39], [40], [41], [42]. Attention mechanisms, such as the self-attention mechanism employed in models like attention U-Net and DANet, enhance feature representation by allowing the network to focus on the most pertinent parts of the image, thereby improving segmentation accuracy. By dynamically weighting the importance of different spatial regions and feature channels, attention mechanisms capture long-range dependencies and contextual information more effectively than traditional convolutional layers. This leads to more precise and context-aware segmentation maps, which are particularly beneficial for complex and cluttered RSIs. The incorporation of attention mechanisms has resulted in notable performance improvements, further establishing CNNs as powerful tools for geospatial analysis. Beyond attention-enhanced CNNs, transformers have emerged as a promising architecture for semantic segmentation, introducing new capabilities and efficiencies [43], [44], [45], [46].

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Transformers have recently revolutionized the semantic segmentation of RSIs. Models such as the Vision Transformer (ViT) [47], [48], Swin Transformer (ST) [49], SegFormer [50], and Segmenter [51] utilize self-attention mechanisms to model long-range dependencies and capture global context. These models approach image segmentation by treating images as sequences of patches, thereby offering greater flexibility and scalability. Transformers excel in managing the high variability and complexity inherent in RSIs, achieving superior performance in terms of accuracy and generalization across diverse datasets [52], [53]. However, Zhang et al. [54] point out that both CNNs and transformers predominantly focus on learning low-frequency components in images, often overlooking high-frequency components.

Current methods frequently struggle to capture fine-grained textures and periodic patterns in RSIs, leading to suboptimal outcomes for complex terrains [55], [56], [57]. Traditional CNN-based architectures excel in capturing spatial hierarchies through encoder–decoder structures, making them highly effective for semantic segmentation. However, they often lack the capacity to fully leverage frequency-domain features, leading to inefficiencies in handling high-frequency details, such as edges and low-frequency components, such as global context. On the other hand, transformers offer enhanced global contextual modeling but suffer from computational inefficiency and limited applicability to high-resolution RSIs. This issue is particularly pronounced in RSIs of varied terrains such as agricultural fields, urban layouts, and natural landscapes. These regions exhibit distinctive textural patterns—forests with varying tree densities, repetitive crop layouts in fields, and grid-like urban structures—critical for accurate land cover classification and environmental monitoring. However, these patterns are not effectively captured by spatial domain models alone. In addition, many land covers, such as croplands or urban grids, display periodic patterns that are better identified in the frequency domain. Spatial models may overlook these patterns due to their focus on local context and spatial correlations, thus failing to leverage the repetitive nature of such textures. Processing RSIs in the frequency domain can complement spatial domain techniques by efficiently capturing both low-frequency components (broad, smooth regions such as forests or water bodies) and high-frequency details (edges and textures of buildings and roads). Utilizing the frequency domain enables better feature differentiation, crucial for achieving accurate and robust segmentation results. In summary, the inability to capture fine-grained textures and periodic patterns results in a misinterpretation of visual features.

1) Traditional neural networks for semantic segmentation primarily rely on spatial features, which may fail to capture underlying patterns and global information in images. This spatial emphasis often leads to suboptimal feature representations, particularly in complex RSIs with diverse and intricate structures.
2) RSIs contain rich spectral information across multiple bands, crucial for accurately distinguishing materials and surface types. Traditional spectral analysis methods often treat all bands equally, leading to inefficiencies. Some bands may carry redundant or less significant information, contributing to sparsity in the spectral representation. Although leveraging sparsity to optimize spectral feature learning is beneficial, there is a lack of methods that effectively integrate sparsity awareness into spectral enhancement.
3) In the frequency domain, image information are distributed across various frequency bands. High-frequency components contain edge and texture details, while low-frequency components represent general structures and smooth areas. A key challenge is effectively separating and optimizing these components, as each requires different handling to positively contribute to segmentation tasks.

Recently, deep learning models have increasingly explored frequency-domain techniques to address the limitations of spatial-only approaches. Methods such as FcaNet [56] and FsaNet [54] leverage frequency channel attention to preserve fine textures and improve segmentation accuracy. For example, FsaNet applies frequency self-attention to enhance edge preservation while reducing computational complexity. Moreover, models like SSCNet [58] and SSCBNet [59] have demonstrated that combining spectral and spatial contexts through joint attention mechanisms can significantly improve the segmentation of RSIs by distinguishing subtle land cover differences that spatial models often overlook. These advancements underscore the importance of frequency-domain learning in achieving state-of-the-art performance in RSI segmentation. To address these challenges, we propose frequency decoupling network (FDNet), a novel architecture built on the CNN-based U-Net framework, enhanced by hierarchical embedding of frequency-domain modules. FDNet introduces three core components: the sparse-aware spectral enhancement module (SSEM), the frequency decoupling attention module (FDAM), and the attentive frequency context module (AFCM). These modules decouple and independently optimize high-frequency and low-frequency features, enabling FDNet to combine the strengths of CNNs' spatial feature learning with frequency-domain contextual refinements. This dual-domain optimization bridges the gap between local feature extraction and global modeling, significantly improving segmentation performance in complex RSI scenarios.

1) We introduce an SSEM that incorporates sparsity awareness into the spectral enhancement process, focusing on the most informative spectral bands and compressing redundant information in the frequency domain. Specifically, SSEM collects the channelwise sparsity and covariance correlations to recalibrate channel weights.
2) We propose a novel FDAM to individually refine high-frequency and low-frequency components of learned representations. The high-frequency refinement ensures accurate edge and detail identification, crucial for precise segmentation boundaries, while optimized low-frequency information provides a strong foundation for understanding overall image structure and context.
3) We design an AFCM that sequentially deploys SSEM and FDAM to enrich spectral context before modeling high-frequency and low-frequency components. We also develop an FDNet by hierarchically embedding AFCMs at each stage. Extensive experiments on four representative datasets demonstrate the superior performance of

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the proposed FDNet compared with several state-of-the-art methods. In addition, ablation experiments assess the impacts of AFCM, SSEM, and FDAM.

The organization of this article is as follows. Section II reviews the literature on semantic segmentation of RSIs and learning in frequency domain. Section III details our proposed FDNet. Section IV discusses results. Section V draws the conclusion of this article.

II. Related Works

A. Semantic Segmentation of RSIs

Semantic segmentation of RSIs involves labeling each pixel in satellite or aerial imagery with land cover or land use types [60], [61], [62], [63], [64], [65]. Due to the extensive geospatial coverage and limited spatial resolution of RSIs, neural networks tailored for specific ground targets often prove unsuitable for these images. In addition, variations in spectral, spatial, and temporal domains introduce ambiguity in semantic labeling [66], [67]. Therefore, developing generic models that can accommodate the unique properties and diverse scenarios of RSIs is essential [68], [69], [70].

ResUNet-a [71], an extension of U-Net, incorporates residual connections and attention modules to enhance feature extraction and context modeling in RSI segmentation. This model combines several advanced computer vision techniques, highlighting the effectiveness of integrating various methods for RSI interpretation. LANet [72] introduces a patchwise attention module to preserve local details before multilevel fusion. Similarly, Li et al. [73] proposed using spatial attention for shallow layers and channel attention for deep layers in the semantic segmentation of satellite images. Following the hybrid design approach, SCAttNet [74], HMANet [75], and HCANet [76] capture and integrate channel and spatial attentional contexts before inference. In addition, Ding et al. [44] developed WiCoNet, which employs two CNNs to extract features at local and global image levels, effectively capturing long-range dependencies to enhance RSI semantic segmentation. Li et al. [77] introduced a synergistic attention model (SAM) that simultaneously captures spatial and channel dependencies, addressing the attention bias in the semantic segmentation of RSIs. Consequently, the constructed SAPNet achieves state-of-the-art performance by enhancing feature representation with comprehensive contextual information.

Recently, transformers have emerged as powerful tools in computer vision, efficiently capturing contextual dependencies in images. SETR [78] frames semantic segmentation as a sequence-to-sequence prediction task, encoding images as a series of patches, and achieves competitive performance on natural images. Similarly, the Segmenter model [51] constructs a segmentation transformer based on a ViT. Before fine-tuning on the semantic segmentation task, the Segmenter model is pre-trained on an image classification dataset. Class masks are then generated using a simple pointwise linear decoder. For RSIs, Blaga and Nedevschi [79] proposed a transformer-based U-Net model with guided focal-axial attention for semantic segmentation of RSIs. This approach provides significant improvements by leveraging both the global attention mechanism of transformers and localized attention within high-dimensional feature maps, making it highly relevant to applications involving high-resolution remote sensing data. Moreover, Wang et al. [80] presented UNetFormer, a U-Net-like transformer model designed to efficiently handle semantic segmentation of urban scene imagery in remote sensing. By incorporating transformer layers within a U-Net architecture, UNetFormer effectively captures both local spatial details and long-range global dependencies. Lin et al. [81] introduced the ST Segmentor, which combines STs with CNN-based architectures to capture both local and global features. Their model refines boundary information in RSIs, showing significant improvements in segmentation accuracy. Likewise, Li et al. [82] introduced GPINet, a hybrid network that integrates CNN and transformer features with geometric priors to enhance the semantic segmentation of high-resolution RSIs. Likewise, He et al. [83] proposed ST-UNet, which combines an ST and a CNN-based U-Net to improve RSI semantic segmentation by leveraging global context and detailed spatial features. Long et al. [84] proposed CLCFormer, a cross-learning network that integrates fine-grained spatial details with long-range global contexts using a combination of CNNs and transformers, achieving state-of-the-art performance on several VHR remote sensing datasets.

CNN-based methods, such as U-Net, excel in learning local features and spatial hierarchies through their encoder–decoder structures and skip connections, making them a popular choice for semantic segmentation tasks. However, their reliance on spatial-domain features alone often limits their ability to capture global dependencies or handle complex textural patterns in RSIs. Although attempts have been made to enhance CNNs with multiscale feature fusion and attention mechanisms, these improvements still do not fully address the need for frequency-domain optimization. Transformer-based methods, by contrast, provide global contextual modeling through self-attention mechanisms, which are particularly advantageous in handling large spatial variations. However, their high computational cost and memory inefficiency make them less practical for high-resolution RSIs.

In the context of RSIs, low-frequency components are useful for recognizing larger areas such as agricultural fields, forests, and water bodies. These components provide a general context that aids in identifying land cover types over broader regions. High-frequency components are crucial for identifying detailed features, such as building edges, road boundaries, and fine textures found in urban areas. These details are essential for distinguishing man-made structures from natural features. However, from the perspective of digital signal processing, CNNs and transformers are both sensitive to low-frequency components, such as broader structures and smooth variations [54]. The joint utilization of spectral and spatial contexts remains suboptimal with existing methods. Therefore, we propose refining features in the frequency domain with specific focus on high frequencies and low frequencies, respectively.

B. Learning in Frequency Domain

In recent years, there has been a growing focus on frequency domain learning in image processing [56], [85].

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John Xu et al. [86] conducted a theoretical evaluation of neural networks' spectral bias using Fourier analysis, highlighting their tendency toward low-frequency functions and difficulties in capturing high-frequency details. This insight spurred subsequent research focusing on high-frequency elements. Azad et al. [87] enhanced the local feature capacity of transformer models through a redesigned self-attention map, decomposing features into the frequency domain to reveal finer details for enriched contextual cues. Gueguen et al. [88] utilized frequency domain features for image classification, while research by Ehrlich and Davis [89] focused on converting spatial-domain CNN models into the frequency domain. Another approach involves selecting frequency channels with SE-blocks to bypass complex conversion processes, leading to the development of a frequency channel attention network that explicitly processes spectral information less apparent in the spatial domain [90]. Specifically, Zhang et al. [54] presented FsaNet, a novel semantic segmentation framework utilizing frequency self-attention to improve edge preservation and reduce computational complexity. FsaNet significantly enhances segmentation performance and efficiency, achieving state-of-the-art results on several benchmark datasets.

In the context of RSIs, by applying techniques like the discrete cosine transform (DCT) or Fourier analysis, models can decouple high-frequency (texture and edges) and low-frequency (smooth regions) components, leading to more nuanced feature extraction. Su et al. [91] developed a complete frequency channel attention network (CFCANet) capable of handling noisy RS images without filtering. CFCANet selectively integrates low-frequency information with feature maps, assigning a discrete cosine transformation frequency component to each original feature map and extracting the most significant eigenvalue per channel. This network demonstrates superior noise resistance by leveraging low-frequency image information. For semantic segmentation, Li et al. [58] introduced the spectrum-space collaborative network (SSCNet), which captures both spectral and spatial dependencies to improve RSI segmentation quality. SSCNet features a joint spectral–spatial attention module (JSSA) employing both spectral attention (SpeA) and spatial attention (SpaA). Furthermore, Li et al. [59] presented SSCBNet, a novel network that integrates spectral and spatial contexts through synergistic attention and cross-fusion modules for the semantic segmentation of RSIs. A recent approach by Zhou et al. [92] introduces a spatial–frequency information integration network (SFINet), which integrates both spatial and frequency domain information for multimodal image fusion tasks like pan-sharpening and depth super-resolution. By leveraging invertible neural operators in the spatial domain and deep Fourier transformation in the frequency-domain, SFINet effectively captures local and global information, achieving superior performance in reconstructing high-frequency details. Likewise, the fourier frequency-domain convolutional neural network (FFDCNet) [93] introduces a frequency-domain learning approach for remote sensing crop classification by applying dynamic frequency filters to enhance interclass differences and reduce intraclass variations. By separating feature maps into low- and high-frequency components, FFDCNet achieves higher classification accuracy and robustness, particularly in addressing issues such as incomplete extractions and fragmented boundaries in crop classification tasks. For change detection, the MFGFNet, proposed by Yuan et al. [94], introduces a multiscale remote sensing change detection network that leverages global filters in the frequency domain to enhance edge preservation and sharpen boundaries in change regions. By combining multiscale CNN feature extraction and frequency-based global filters, MFGFNet achieves superior performance on change detection datasets, offering a new perspective for frequency-domain processing.

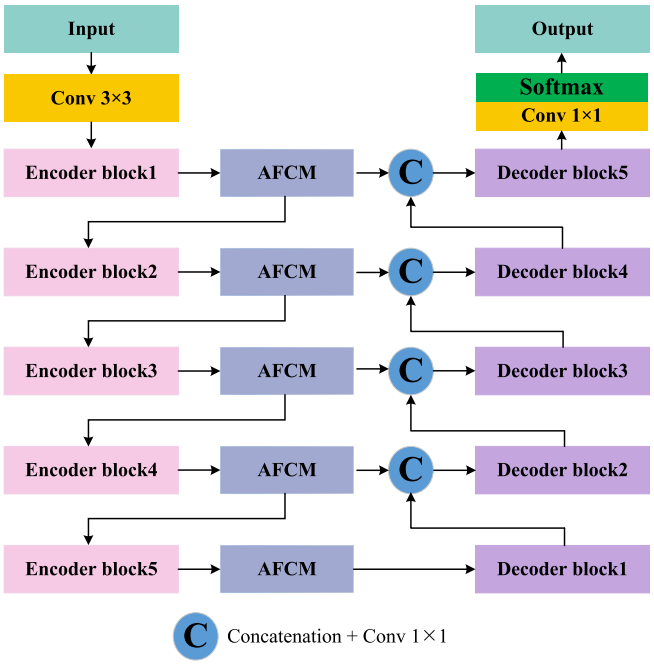


Fig. 1. Overview of the proposed FDNet. The architecture is based on a U-Net framework with a symmetric encoder–decoder structure and skip connections, enhanced by the hierarchical integration of AFCMs. These modules independently refine high-frequency and low-frequency features, allowing the network to achieve multiscale optimization. This figure illustrates how FDNet combines spatial and frequency-domain processing to enhance segmentation performance in RSIs.

In summary, spectral analysis in the frequency domain is beneficial for distinguishing subtle spectral signatures of different land cover classes, enabling the model to differentiate between features with similar spatial but distinct spectral properties. Therefore, leveraging frequential features for RSIs is an effective method to exploit inherent spectral attributes. This approach not only enhances differentiation between land cover classes but also improves model generalization and facilitates the integration of spectral context in semantic segmentation, resulting in more accurate and robust outcomes in remote sensing applications.

III. Method

A. Overview of the Proposed FDNet

Fig. 1 provides an overview of the proposed FDNet, which incorporates a hierarchical structure embedding AFCMs across multiple stages. This architecture facilitates multiscale feature refinement, enabling accurate segmentation of objects with varying sizes and complexities in RSIs. FDNet adopts a hierarchical structure based on ResNet101 [95], incorporating symmetric encoder–decoder blocks and skip connections akin

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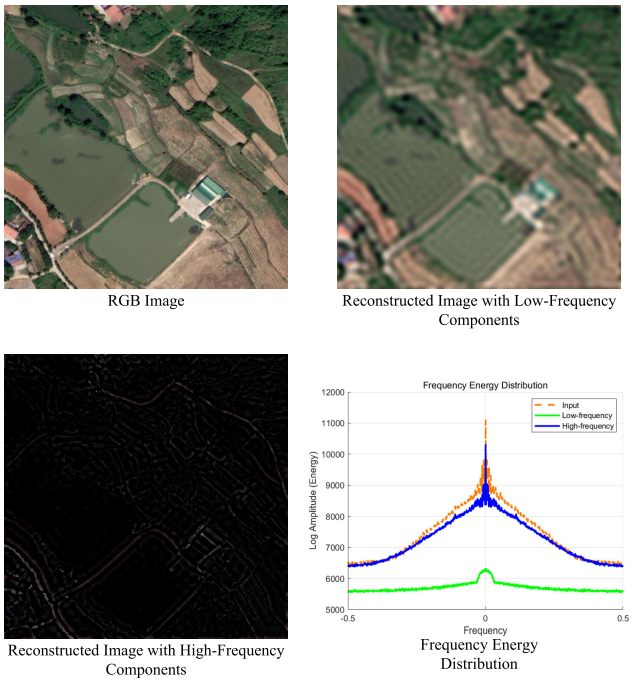


Fig. 2. Understanding images in the frequency domain.

to U-Net. By integrating frequency-domain modules (SSEM, FDAM, and AFCM), FDNet enhances the capabilities of CNN-based architectures by independently refining high- and low-frequency components. This enables FDNet to bridge the gap between the spatial feature learning of CNNs and the contextual modeling of transformers.

Specifically, the hierarchical embedding of AFCMs enables FDNet to achieve refined multiscale feature representation, significantly enhancing the network's robustness and generalization performance. By jointly leveraging spatial and frequency domain processing, FDNet effectively mitigates challenges such as noise, texture variability, and resolution inconsistencies, which are prevalent in RSI segmentation. This dual-domain optimization strategy not only preserves critical edge details but also ensures coherent global context, making FDNet well-suited for diverse imaging scenarios and complex environments. Consequently, FDNet demonstrates superior adaptability and precision across varying spatial resolutions and challenging terrain types, establishing its efficacy in real-world applications.

B. Understanding Images in the Frequency Domain

In the realm of signal processing theory, decomposing an image into its frequency components provides a clear understanding of the image's information content. The upper left of Fig. 2, the red, green, and blue (RGB) format of an RSI, showcases a landscape with fields and water bodies. Upon decomposing this image into high-frequency (bottom left of Fig. 2) and low-frequency (upper right of Fig. 2) components, distinct characteristics emerge. The low-frequency image emphasizes broad, smooth areas and general shapes, indicating the overall structure and layout of the scene. In contrast, the high-frequency image highlights edges and fine details, such as the borders of fields and smaller structures.

The frequency energy distribution (bottom right of Fig. 2) further corroborates these observations. The input image's spectrum shows a significant concentration of energy in the low-frequency range, indicating the dominance of large-scale features and smooth transitions. However, there is also a considerable presence of high-frequency energy, corresponding to the detailed textures and sharp edges. The low-frequency spectrum, centered around zero frequency, confirms the presence of broad, smooth areas with minimal high-frequency content. Conversely, the high-frequency spectrum displays a pronounced distribution at higher frequencies, aligning with the intricate features observed in the high-frequency image.

These findings underscore the importance of frequency-domain decomposition in deep neural networks for feature optimization. By separating high-frequency and low-frequency information, we can independently model and optimize contextual features at different scales, as it enables more effective learning of diverse feature sets, enhancing the network's ability to capture both broad structural information and fine-grained details. Thus, frequency-domain decoupling for contextual modeling is not only necessary but also highly beneficial and grounded in established signal processing principles.

C. Details of AFCM

As shown in Fig. 3(a), we first project the input feature maps into the frequency domain by a 2-D discrete cosine transformation (2D-DCT). Formally,

$$F_f(u,v,c) = \alpha(u)\alpha(v) \times \sum_{x=0}^{H-1}\sum_{y=0}^{W-1} F_{\text{in}}(x,y,c) \times \cos\left(\frac{\pi(2x+1)u}{2H}\right)\cos\left(\frac{\pi(2y+1)v}{2W}\right) \quad (1)$$

where $u = 0, 1, \ldots, H-1$ and $v = 0, 1, \ldots, W-1$, $c = 1, 2, \ldots, C$, and $\alpha(u)$ and $\alpha(v)$ are the normalization factors defined as follows:

$$\alpha(k) = \begin{cases} \sqrt{\frac{1}{N}}, & \text{if } k = 0 \\ \sqrt{\frac{2}{N}}, & \text{if } k \neq 0 \end{cases} \quad (2)$$

where N is the dimension size [i.e., H or W].

Then, an SSEM is applied to enrich the informative spectral bands in the frequency domain. Subsequently, FDAM is utilized to model frequency domain context for refining learned representations. By element summation with F_f and 2-D inverse DCT (2D-iDCT), we reproject the frequency context-refined representations. Sections III-D and III-F will detailedly present SSEM and FDAM.

D. Details of SSEM

As shown in Fig. 3(b), the SSEM stems from the necessity to improve spectral feature representation by leveraging the frequency domain sparsity and covariance matrices. The covariance between channels measures the interdependence and relationships among different channels, providing a global

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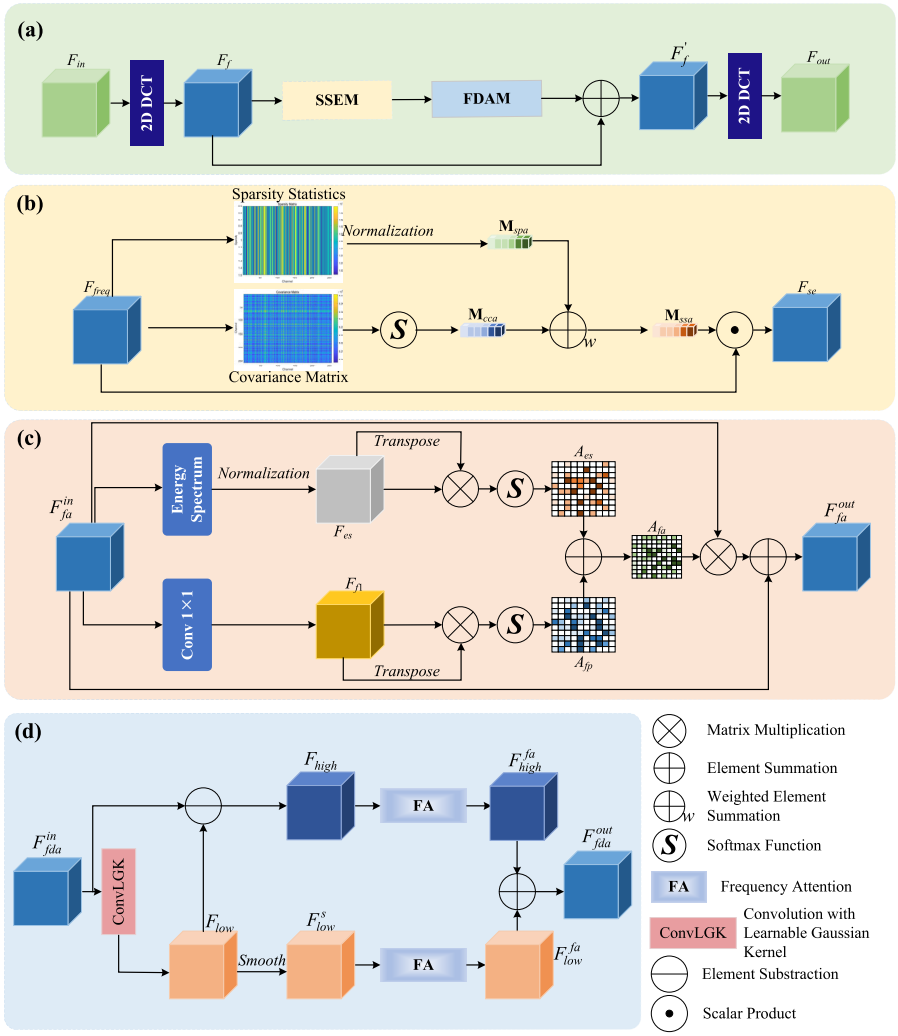


Fig. 3. Details of the proposed FDNet's frequency-decoupling modules. (a) AFCM embeds spectral and spatial refinement within the frequency domain. The 2D-DCT transforms input features into frequency components, which are enhanced by SSEM and FDAM. (b) SSEM highlights significant spectral bands using sparsity and covariance matrices to recalibrate feature channels. (c) FA refines frequency-domain features by aligning energy spectrum and positional attention. (d) FDAM separates high- and low-frequency components, refining each to preserve edges and contextual smoothness.

perspective on channel correlations. By calculating the covariance matrix, we can capture the linear relationships between different channels, which is vital for understanding and utilizing global information in frequency-domain features. Conversely, the sparsity matrix reflects the sparsity level of each channel in the frequency domain. High-sparsity channels typically contain more significant information or prominent features, as they have fewer nonzero elements, each potentially carrying high signal strength. Integrating these two aspects, the SSEM aims to identify and amplify important frequency bands while reducing redundancy, thus preserving the main characteristics of the signal.

Given the input frequency-domain feature map $\mathbf{F}_{\text{freq}}$ of dimensions $H \times W \times C$, we aim to compute the sparsity matrix $\mathbf{M}_{\text{spa}}$, which represents the normalized sparsity for each channel. The resulting $\mathbf{M}_{\text{spa}}$ has dimensions $C \times 1$. For each channel c, count the number of nonzero elements in $\mathbf{F}_{\text{freq}}$. This gives the sparsity measure for each channel

$$S_c = \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} \delta\left(\mathbf{F}_{\text{freq}}(x, y, c) \neq 0\right). \tag{3}$$

Here, $\delta(\cdot)$ is the indicator function that equals 1 if the condition is true, and 0 otherwise. Then, we normalize the sparsity values across all channels to ensure they sum to 1

$$\mathbf{M}_{\text{spa}}(c) = \frac{S_c}{\sum_{k=1}^{C} S_k}, \quad c = 1, 2, \ldots, C. \tag{4}$$

At the bottom branch, we first form the covariance matrix of frequency-domain feature maps. Initially, the mean vector μ for each channel can be produced as follows:

$$\mu_c = \frac{1}{H \times W} \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} \mathbf{F}_{\text{freq}}(x, y, c) \tag{5}$$

where μ_c represents the mean vector for channel c. Then, we subtract the mean vector from the feature matrix to center it

$$\mathbf{X}_{\text{centered}} = \mathbf{X} - \mu \tag{6}$$

where $\mathbf{X}$ is the reshaped 2-D matrix of dimensions $(H \times W) \times C$, and $\mathbf{X}_{\text{centered}}$ denotes the centered feature matrix. Afterward, covariance matrix Σ of dimensions $C \times C$ can be computed

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as follows:

$$\Sigma = \frac{1}{H \times W - 1} \mathbf{X}_{\text{centered}}^{\text{T}} \mathbf{X}_{\text{centered}}. \tag{7}$$

By extracting the diagonal elements of the covariance matrix, we have the variance vector of each channel, which reflects the importance and variation of individual channels. Finally, the normalized covariance channel attention for the channel is formed after softmax normalization

$$\mathbf{M}_{\text{cca}}(c) = \frac{e^{\sigma_c}}{\sum_{k=1}^{C} e^{\sigma_k}}, \quad c = 1, 2, \ldots, C \tag{8}$$

where σ_c is the variance for channel c and $\mathbf{M}_{\text{cca}}(c)$ denotes the normalized covariance channel attention for channel c.

To leverage both the sparsity information, which highlights significant features with fewer nonzero elements, and the covariance information, which captures interchannel dependencies, the weighted summation, a simple yet effective operation, is utilized

$$\mathbf{M}_{\text{ssa}}(c) = \alpha \mathbf{M}_{\text{spa}}(c) + (1 - \alpha)\mathbf{M}_{\text{cca}}(c), \quad c = 1, 2, \ldots, C \tag{9}$$

where $\mathbf{M}_{\text{ssa}}$ with dimensions of $C \times 1$ represents the sparse-aware spectral attention map and α is a weighting factor that balances the contribution of sparsity and covariance contexts (set as 0.5 in this study). By deploying scalar products, the input feature maps are enhanced by incorporating sparsity-aware spectral context.

E. Details of FA

In the context of attention mechanisms, SE [35] performs channel weighting using global pooling but lacks spatial or frequency considerations. CBAM [36] combines spatial and channel attention but does not decouple frequency components. DAM [37] and SAPM [77] focus on spatial dependencies but do not optimize feature refinement in the frequency domain. Unlike SE, CBAM, and DAM, which operate in spatial or channel domain, frequency attention (FA) explicitly decouples high-frequency (edges and textures) and low-frequency (global context) components, bridging the gap by simultaneously optimizing spatial, channel, and frequency features. Specifically, our proposed FA refines frequency domain features by analyzing their energy spectrum and positional correlations, enhancing global and detailed feature capture. By analyzing the energy spectrum and positional correlation of frequency domain features, FA can better capture both global and detailed information in images. By analyzing the energy spectrum, FA can highlight details and edges in images, enhancing the representation of these fine features. Moreover, FA adapts the weights of different frequency components and positions using a self-attention mechanism, improving the flexibility and accuracy of feature representation.

As shown in Fig. 3(c), given the input frequency-domain feature map $F_{fa}^{\text{in}} \in \mathbb{R}^{C \times H \times W}$, where C, H, and W denote the number of channels, height, and width, respectively, the energy spectrum matrix $E(c, u, v)$ can be described as follows:

$$E(c, u, v) = |F_{fa}^{\text{in}}(c, u, v)|^2 \tag{10}$$

where $E(c, u, v)$ is the energy value at channel c and position (u, v). The normalized energy spectrum matrix $F_{\text{es}} \in \mathbb{R}^{C \times H \times W}$ can be formed as follows:

$$F_{\text{es}}(c, u, v) = \frac{E(c, u, v)}{\sum_{c=1}^{C} \sum_{u=1}^{H} \sum_{v=1}^{W} E(c, u, v)}. \tag{11}$$

Subsequently, the positionwise energy spectrum correlations are computed as follows:

$$A_{\text{es}}(i, j) = \text{Softmax}\left(\sum_{c=1}^{C} F_{\text{es}}(c, i) \cdot F_{\text{es}}(c, j) \right) \tag{12}$$

where $A_{\text{es}} \in \mathbb{R}^{HW \times HW}$ is the positional energy spectrum correlation matrix. Likewise, the bottom branch calculates the standard self-attention weights of each position $A_{fp} \in \mathbb{R}^{HW \times HW}$.

Postfused by a weighted summation, we can obtain the FA map as follows:

$$A_{fa} = \gamma A_{\text{es}} + (1 - \gamma) A_{fp} \tag{13}$$

where γ is the weighting coefficient (set as 0.5 in this study). Finally, the output of FA can be produced by

$$F_{fa}^{\text{out}} = F_{fa}^{\text{in}} \times A_{fa} + F_{fa}^{\text{in}}. \tag{14}$$

In summary, FA simultaneously processes energy spectrum attention and positional attention in parallel, capturing both global and detailed information to improve the accuracy and flexibility of feature representation.

F. Details of FDAM

Fig. 3(d) presents the pipeline of FDAM, which is designed to refine frequency components individually, applying specific strategies to enhance high-frequency details such as edges and textures, while also optimizing low-frequency components for structural and contextual accuracy. This frequency decoupling approach allows the model to adaptively adjust the weights of different frequency components, thereby improving the robustness of feature representation.

Given the input frequency feature map $F_{\text{fda}}^{\text{in}} \in \mathbb{R}^{C \times H \times W}$, where C, H, and W denote the number of channels, height, and width, respectively, we first deploy a convolution with learnable Gaussian kernel (ConvLGK) [96] to separate the low-frequency components as $F_{\text{low}} = \text{ConvLGK}(F_{\text{fda}}^{\text{in}})$. Then, we have the high-frequency feature map by

$$F_{\text{high}} = F_{\text{fda}}^{\text{in}} - F_{\text{low}}. \tag{15}$$

In parallel, we further apply Gaussian smoothing to the low-frequency feature matrix F_{low} to obtain the smoothed low-frequency feature matrix F_{low}^{s}. Subsequently, two FAs are implemented to F_{low}^{s} and F_{high} individually. With the element summation, the output feature of FDAM can be represented as follows:

$$F_{\text{fda}}^{\text{out}} = \text{FA}\left(F_{\text{high}}\right) + \text{FA}\left(F_{\text{low}}^{s}\right) \tag{16}$$

where FA(.) is presented in Section III-E.

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G. Proofs of Using Learnable Gaussian Kernel Convolution

This section derives the gradient propagation of the learnable Gaussian kernel convolution to establish its theoretical feasibility as a low-frequency feature extractor. Unlike ordinary convolution, Gaussian kernel convolution dynamically adjusts the frequency distribution of features through learnable parameters, enabling enhanced low-frequency feature extraction and improved semantic segmentation performance. A Gaussian convolutional kernel is typically used for image smoothing and noise reduction [97]. A 2-D Gaussian convolutional kernel can be represented as follows:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \tag{17}$$

where $G(x, y)$ is the value of the Gaussian kernel at position (x, y) and σ is the standard deviation of the Gaussian distribution, controlling the degree of smoothing. To make the Gaussian convolutional kernel learnable during training, we set the standard deviation σ as a learnable parameter

$$G_\theta(x, y) = \frac{1}{2\pi\theta^2} \exp\left(-\frac{x^2 + y^2}{2\theta^2}\right) \tag{18}$$

where θ is the learnable parameter representing the standard deviation.

Using the learnable Gaussian convolution kernel G_θ to perform convolution on the input feature map, we extract the low-frequency features:

$$F_{\text{low}}(i, j) = \sum_{m=-k}^{k} \sum_{n=-k}^{k} G_\theta(m, n) \cdot F_{\text{fda}}^{\text{in}}(i + m, j + n) \tag{19}$$

where $F_{\text{low}}(i, j)$ is the low-frequency feature value at position (i, j), $F_{\text{fda}}^{\text{in}}$ is the input feature map, and k is the radius of the convolutional kernel.

During network training, the parameters of the convolution kernel θ are updated through backpropagation. The specific update formula is as follows:

$$\theta \leftarrow \theta - \eta \frac{\partial L}{\partial \theta} \tag{20}$$

where η is the learning rate, L is the loss function, and $(\partial L/\partial\theta)$ is the gradient of the loss function with respect to the parameter θ.

To perform backpropagation, we need to compute the gradient of the loss function L with respect to the convolutional kernel parameter θ. According to the chain rule, the gradient calculation is as follows:

$$\frac{\partial L}{\partial \theta} = \sum_i \sum_j \frac{\partial L}{\partial F_{\text{low}}(i, j)} \cdot \frac{\partial F_{\text{low}}(i, j)}{\partial \theta} \tag{21}$$

where $(\partial L/\partial F_{\text{low}}(i, j))$ is the gradient of the loss function with respect to the low-frequency feature map $F_{\text{low}}(i, j)$, and $(\partial F_{\text{low}}(i, j)/\partial\theta)$ is the gradient of the low-frequency feature map with respect to the Gaussian kernel parameter θ.

The gradient of the low-frequency feature map with respect to the Gaussian kernel parameter θ is calculated as follows:

$$\frac{\partial F_{\text{low}}(i, j)}{\partial \theta} = \sum_{m=-k}^{k} \sum_{n=-k}^{k} \frac{\partial G_\theta(m, n)}{\partial \theta} \cdot F_{\text{fda}}^{\text{in}}(i + m, j + n). \tag{22}$$

The gradient of the Gaussian kernel with respect to the parameter θ is calculated as follows:

$$\frac{\partial G_\theta(m, n)}{\partial \theta} = G_\theta(m, n) \cdot \left(\frac{x^2 + y^2}{\theta^3} - \frac{2}{\theta}\right). \tag{23}$$

Ordinary convolution can also extract frequency domain features through the composition of kernels. However, its frequency-domain behavior relies on fixed kernel designs. In contrast, our Gaussian kernel convolution introduces learnable parameters that enable dynamic optimization of low-frequency feature extraction, providing a significant extension to conventional methods. While ordinary convolution can extract frequency-domain features, its frequency response during gradient propagation is fixed. In contrast, our proposed learnable Gaussian kernel convolution dynamically adjusts its parameters to achieve precise control over frequency components, particularly low-frequency optimization. This design has been demonstrated to outperform traditional methods in enhancing segmentation performance.

In summary, the Gaussian kernel convolution differentiates itself from traditional convolution methods through its learnable parameterization, which enables adaptive frequency optimization. Standard convolutions are limited by static kernel designs, whereas the Gaussian kernel convolution achieves superior feature refinement by tailoring its response to low-frequency characteristics during training. The empirical results throughout our experiments demonstrate the practical benefits of this capability in achieving improved segmentation performance, without requiring additional computational complexity compared with advanced convolutional variants.

H. Loss Function

In semantic segmentation, the conventional cross-entropy loss primarily focuses on measuring the classification error between the predicted outputs and the ground truth in the spatial domain. However, in applications such as remote sensing, frequency domain information (such as spectral features) is equally important. By introducing the frequency-domain SSIM loss alongside the cross-entropy loss, we create a composite loss function that ensures the model excels in the spatial domain while also preserving and utilizing critical features in the frequency domain.

The loss function is expressed as follows:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{CE}} + \beta \mathcal{L}_{\text{freq-SSIM}} \tag{24}$$

where $\mathcal{L}_{\text{CE}}$ represents the cross-entropy loss in the spatial domain, which measures the pixelwise classification error, $\mathcal{L}_{\text{freq-SSIM}}$ denotes the SSIM loss in the frequency domain, which evaluates the structural similarity between the predicted output and the ground truth in the frequency domain, and α and β are weight parameters that balance the contribution of each loss component. Notably, in this study, both α and β are set as 0.5 to balance the contributions of the $\mathcal{L}_{\text{CE}}$

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and $\mathcal{L}_{\text{freq-SSIM}}$. This equal weighting ensures that both losses have a similar influence on model optimization, preventing one loss from overwhelming the other during training. A preliminary analysis of the magnitudes of the two losses shows that they are generally of comparable scale throughout the training process. To further validate this choice, we conducted sensitivity analyses by varying the ratio of α and β and found that the equal weighting configuration led to the most stable and effective performance across multiple tasks. While task-specific tuning of these weights may be possible, the balanced setting provides a practical and efficient default.

Cross-Entropy Loss $\mathcal{L}_{\text{CE}}$ in the spatial domain is computed using the formula

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[Y_i \log\left(\hat{Y}_i\right) + (1 - Y_i)\log\left(1 - \hat{Y}_i\right)\right] \quad (25)$$

where N is the total number of pixels in the image, Y_i is the ground-truth label for the ith pixel, and $\hat{Y}_i$ is the predicted probability for the ith pixel.

In the context of *frequency-domain SSIM loss* $\mathcal{L}_{\text{freq-SSIM}}$, we first convert the predicted output $\hat{Y}$ and the ground truth Y into the frequency domain by computing their DCTs and obtaining the amplitude spectra

$$\mathcal{F}(Y) = \text{DCT2}(Y), \quad \mathcal{F}\left(\hat{Y}\right) = \text{DCT2}\left(\hat{Y}\right) \quad (26)$$

$$\mathcal{A}(Y) = |\mathcal{F}(Y)|, \quad \mathcal{A}\left(\hat{Y}\right) = |\mathcal{F}\left(\hat{Y}\right)|. \quad (27)$$

Next, we compute the SSIM value based on the amplitude spectra and define the frequency-domain SSIM loss as follows:

$$\mathcal{L}_{\text{freq-SSIM}} = 1 - \text{SSIM}\left(\mathcal{A}(Y), \mathcal{A}\left(\hat{Y}\right)\right). \quad (28)$$

Here, SSIM stands for structural similarity index, which measures the similarity in structure between two images. The final combined loss $\mathcal{L}$ is obtained by linearly weighting the cross-entropy loss and the frequency-domain SSIM loss. This combined loss function is optimized during the training process through backpropagation, ensuring the model's effectiveness in both the spatial and frequency domains.

IV. Experiments and Results

A. Settings

The proposed FDNet and the comparative methods were implemented using PyTorch on a Linux operating system, utilizing an NVIDIA A40 GPU. Table I outlines the experimental settings. We selected 12 methods for comparison. DeepLab V3+ [33] and DANet [37] were used as baselines for the semantic segmentation task. For RSI-specific models, we reproduced RAANet [98], MACU-Net [99], SCAttNet [74], A2FPN [100], LANet [72], and transformer-based methods, like ST-UNet [83] and CLCFormer [84]. In addition, four frequency-domain-based methods, namely, FcaNet [54], FsaNet [56], SSCNet [58], and SSCBNet [59], were also included in the comparison.

TABLE I
Experimental Settings

Items	Settings
Learning strategy	SGD
Initial learning rate	0.02
Loss function	Cross-entropy
Max epoch	500
GPU memory	48 GB
Batch size	32

B. Datasets

1) ISPRS Vaihingen Dataset: The ISPRS Vaihingen dataset [101], derived from an aerial platform, offers very high-resolution imagery at 9 cm per pixel. It comprises 33 images, each with dimensions approximately 2494 × 2064 pixels, and utilizes the RGB spectral bands. The entire dataset is segmented into six categories based on pixels: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background.

2) ISPRS Potsdam Dataset: The Potsdam dataset [102] consists of 38 orthophotos, each with a ground sampling distance (GSD) of 5 cm and a size of 6000 × 6000 pixels. Each image includes four bands: near-infrared (IR), R, G, and B, along with the corresponding digital surface model (DSM) and normalized DSM (NDSM) data. The classification categories in this dataset are identical to those in the Vaihingen dataset. In the experiments conducted in this study, only the RGB image data were used for training and testing.

3) LoveDA Dataset: The LoveDA dataset [103] introduces a significant challenge in the semantic segmentation of large-scale satellite images with a spatial resolution of 0.3 m. Acquired from the Google Earth Platform, LoveDA covers an area exceeding 536 square kilometers, incorporating both rural and urban regions of three cities: Nanjing, Changzhou, and Wuhan. Each image in the dataset has a spatial size of 1024 × 1024 pixels. For this study, we utilized 2522 images for training, 834 images for validation, and 835 images for testing. The class distribution is imbalanced, and similar objects differ in scale, size, and surface type, making LoveDA an exceptionally challenging dataset.

4) DeepGlobe Dataset: The DeepGlobe Land Cover Classification dataset [104] comprises high-resolution satellite images provided by DigitalGlobe. This dataset includes 803 images, each with a spatial size of 2448 × 2448 pixels. The corresponding well-annotated ground-truth images categorize the data into seven classes: urban land, agricultural land, rangeland, forest land, water, barren land, and unknown areas. For three different datasets, we cropped the dataset into subpatches (a subpatch is with a spatial size of 256 × 256) and divided them into training, validation, and testing sets in a 1:1:1 ratio.

C. Evaluation Metrics

We employ four metrics to evaluate the performance of the predicted results on the test set: classwise $F1$-score ($F1$), average $F1$ (AF) score across all classes, overall accuracy (OA), and mean intersection over union (mIoU). The equations for $F1$, OA, and IoU are provided in the following equations, respectively:

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (29)$$

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TABLE II
RESULTS ON THE ISPRS VAIHINGEN DATASET. CLASSWISE F_1-SCORE, AF, OA, AND mIoU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU
DeepLab V3+ [33]	87.99	87.80	72.72	85.55	47.37	76.29	74.47	69.69
DANet [37]	91.57	90.37	80.01	88.15	58.50	79.32	77.50	72.46
RAANet [99]	90.09	89.72	76.22	86.13	48.55	78.14	75.33	70.49
MACU-Net [100]	89.52	90.11	74.32	87.15	62.70	80.76	77.86	72.86
SCAttNet [75]	90.44	91.63	81.22	81.49	71.54	83.26	81.52	75.29
A2FPN [101]	92.99	95.17	80.98	89.25	87.62	89.20	87.07	81.06
LANet [73]	93.81	95.70	82.93	91.94	88.44	90.56	88.48	81.72
ST-UNet [84]	91.52	92.72	82.19	82.46	72.39	84.26	82.29	76.01
FcaNet [58]	89.52	90.11	74.32	87.15	62.70	80.76	77.86	72.86
CLCFormer [85]	90.01	92.26	77.17	87.97	80.08	85.50	83.46	78.10
FsaNet [54]	93.81	95.70	82.93	91.94	88.44	90.56	88.48	81.72
SSCNet [59]	93.05	95.94	83.14	92.17	88.66	90.59	88.61	81.91
SSCBNet [60]	93.09	95.52	82.99	92.67	90.11	90.88	88.89	82.87
FDNet (ours)	**93.92**	**96.56**	**84.33**	**93.41**	**92.05**	**92.05**	**91.05**	**84.09**

$$\mathrm{OA} = \frac{(\mathrm{TP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})} \tag{30}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})}. \tag{31}$$

Here, precision and recall are defined as follows:

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{32}$$

$$\mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{33}$$

where TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative samples, respectively.

D. Comparative Experiments

1) Results on the ISPRS Vaihingen Dataset: The performance of FDNet was evaluated on the ISPRS Vaihingen dataset, and its results were compared with several state-of-the-art models, as shown in Table II. FDNet consistently outperforms competing methods across all key metrics, including classwise F_1-score, AF, OA, and mIoU. In particular, FDNet excels in challenging categories like "car" with an F_1-score of 92.05, significantly improving the segmentation of small, fine-grained objects. The model also demonstrates strong performance in larger classes such as "impervious surfaces" (93.92) and "building" (96.56), highlighting its superior ability to distinguish between complex urban structures.

FDNet achieves an AF of 92.05 and an OA of 91.05%, which represent notable improvements over the baseline methods, including LANet and SSCBNet, which achieve AF values of 90.56 and 90.88, respectively. Similarly, the OA of models such as A2FPN (87.07%) and SSCNet (88.61%) remains significantly lower than that of FDNet. When compared with transformer-based methods, FDNet demonstrates superior performance across all key metrics. Specifically, FDNet achieves the highest F_1-score for the "car" class (92.05) and excels in challenging categories such as "impervious surfaces" (93.92) and "building" (96.56), which are essential for urban semantic segmentation. This is in contrast to ST-UNet, which records an F_1-score of 72.39 for "car" and 92.72 for "building," and CLCFormer, which achieves slightly lower scores of 80.08 and 92.26 for "car" and "building," respectively. In terms of AF (92.05) and OA (91.05%), FDNet outperforms ST-UNet (84.26% and 82.29%, respectively), and CLCFormer (85.50% and 83.46%, respectively), highlighting its ability to better capture intricate spatial and contextual information, essential for high-resolution RSIs.

In terms of mIoU, FDNet records the highest score of 84.09%, outperforming the best-performing alternatives such as SSCBNet (82.87%) and LANet (81.72%). This demonstrates FDNet's ability to more accurately predict segmentation boundaries and handle overlapping class labels. The model's superior performance is largely driven by its frequency decoupling attention mechanism, which refines high-frequency and low-frequency features separately, ensuring more precise edge detection and boundary refinement.

Based on the visualization comparisons shown in Fig. 4, the visual inspection of results on the ISPRS Vaihingen test set reveals several key insights into the performance of different segmentation methods. The comparison illustrates how various models handle complex urban environments, with particular attention to challenging classes, such as buildings, trees, low vegetation, and cars.

The proposed FDNet [Fig. 4(p)] demonstrates superior segmentation accuracy, particularly in delineating fine details and preserving the structural integrity of buildings and roads, when compared with earlier methods like DeepLab V3+ [Fig. 4(c)] and DANet [Fig. 4(d)]. FDNet's ability to effectively separate closely situated classes, such as buildings and low vegetation, highlights the effectiveness of the frequency decoupling attention mechanism in refining boundary precision and reducing misclassifications. FDNet excels in preserving sharp boundaries, which is crucial for urban segmentation tasks. While models such as SCAttNet [Fig. 4(g)] and A2FPN [Fig. 4(h)] also perform well in certain aspects, they occasionally struggle with smaller object classes like cars, where FDNet shows a noticeable improvement in both accuracy and consistency. This is particularly evident in the "car" class, where FDNet maintains better segmentation precision. In addition, CLCFormer [Fig. 4(l)] shows a strong ability to capture global context and handle large object classes such as buildings and trees. However, it faces challenges in fine-grained segmentation, especially for smaller objects such as

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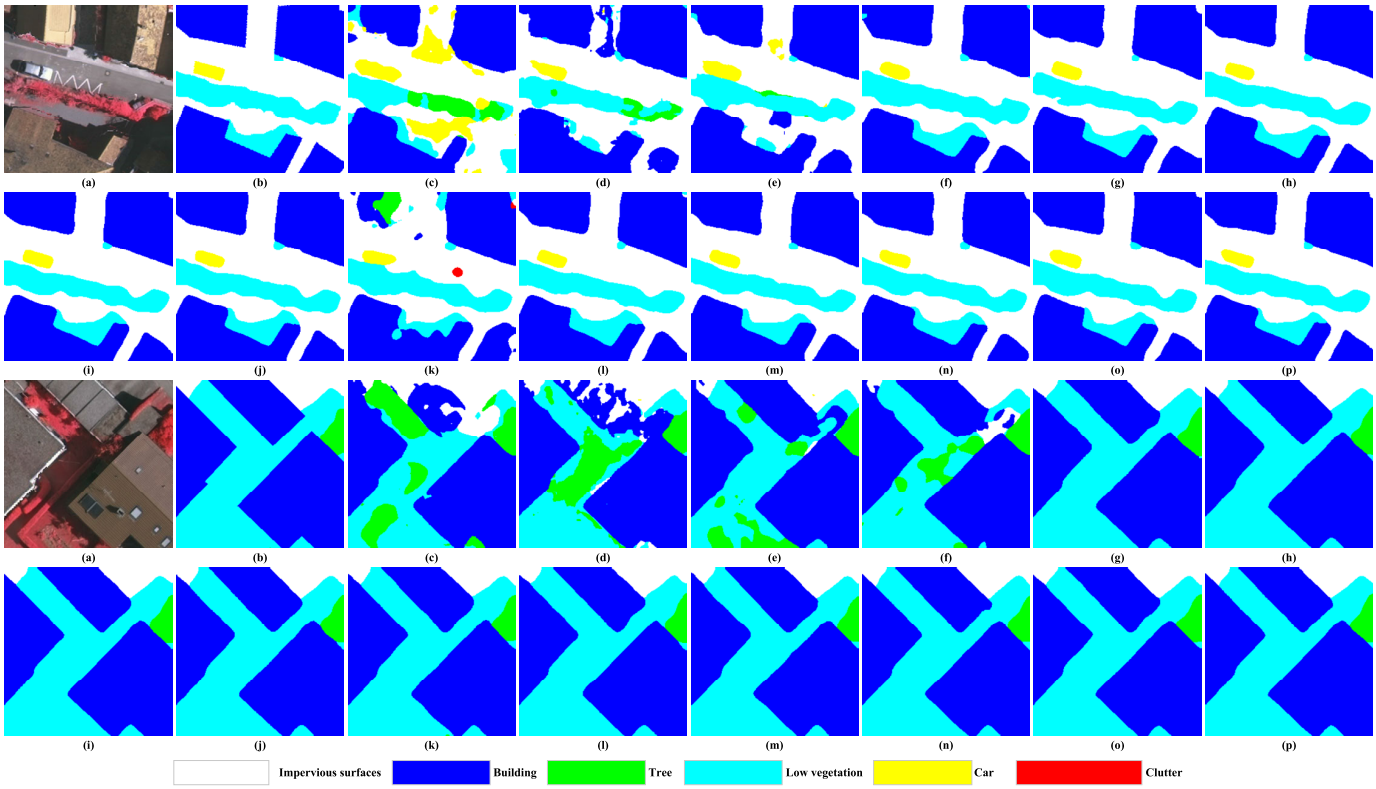


Fig. 4. Visualization comparisons on the ISPRS Vaihingen test set. (a) Input, (b) ground truth, (c) DeepLab V3+, (d) DANet, (e) RAANet, (f) MACU-Net, (g) SCAttNet, (h) A2FPN, (i) LANet, (j) ST-UNet, (k) FcaNet, (l) CLCFormer, (m) FsaNet, (n) SSCNet, (o) SSCBNet, and (p) FDNet (ours).

TABLE III
RESULTS ON THE ISPRS POTSDAM DATASET. CLASSWISE F_1-SCORE, AF, OA, AND MIOU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU
DeepLab V3+ [33]	84.49	86.13	77.37	77.37	85.47	82.17	80.82	73.59
DANet [37]	86.96	92.04	80.14	79.95	89.36	85.69	83.99	77.13
RAANet [99]	90.40	95.70	87.20	81.80	77.60	86.54	85.21	77.95
MACU-Net [100]	89.16	92.76	86.80	83.02	79.25	86.20	85.90	77.99
SCAttNet [75]	91.87	96.90	85.24	87.05	92.78	90.77	89.06	80.83
A2FPN [101]	90.67	95.69	85.54	85.44	91.55	89.78	88.25	80.67
LANet [73]	92.41	96.38	86.82	87.77	93.44	91.36	90.26	81.19
ST-UNet [84]	92.75	97.82	86.05	87.87	93.66	91.63	90.39	81.60
FcaNet [58]	92.16	97.21	85.51	87.32	93.07	91.06	89.34	81.09
CLCFormer [85]	92.66	96.64	87.05	88.00	93.69	91.61	89.97	81.68
FsaNet [54]	93.18	**98.28**	86.46	88.29	94.10	92.06	90.61	81.79
SSCNet [59]	93.19	97.16	89.50	91.41	93.26	92.90	91.03	82.55
SSCBNet [60]	93.52	97.23	87.90	88.61	96.42	92.73	91.31	83.09
FDNet (ours)	**94.11**	97.55	**89.97**	**91.95**	**96.46**	**94.01**	**92.47**	**84.92**

cars, where its performance lags behind that of FDNet. The limitations of CLCFormer in these areas are due to its reliance on broader contextual information, which, while helpful for large-scale structures, can overlook finer details crucial for smaller object classes.

Overall, the visual inspection underscores FDNet's capability to balance detail preservation and class separation, achieving a higher level of segmentation quality that is visually evident and quantitatively supported by the previous performance metrics discussed. Its superior performance in challenging scenarios, especially with fine-grained objects like cars, sets it apart from both CNN-based and transformer-based methods, including CLCFormer.

2) Results on the ISPRS Potsdam Dataset: Table III provides a comprehensive comparison of various state-of-the-art methods on the ISPRS Potsdam dataset. Starting with the baseline methods, DeepLab V3+ demonstrates solid performance across most categories, with an AF of 82.17 and an mIoU of 73.59%. However, despite its effectiveness in handling general segmentation tasks, it exhibits limitations in distinguishing finer details, particularly in the low vegetation and tree categories, where its F_1-scores are notably lower. This suggests that while DeepLab V3+ is robust, it may lack the capacity to fully capture the complexities of high-resolution remote sensing data.

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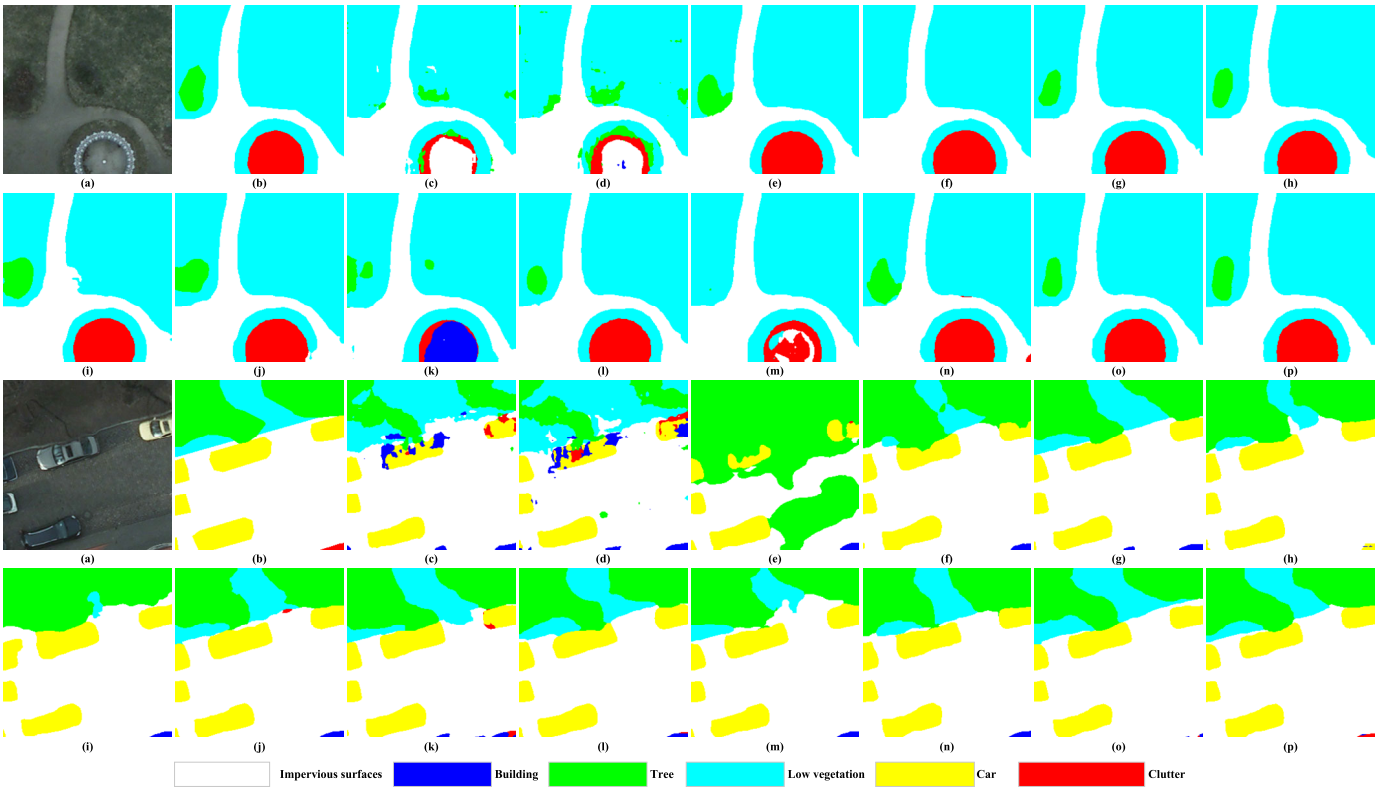


Fig. 5. Visualization comparisons on the ISPRS Potsdam test set. (a) Input, (b) ground truth, (c) DeepLab V3+, (d) DANet, (e) RAANet, (f) MACU-Net, (g) SCAttNet, (h) A2FPN, (i) LANet, (j) ST-UNet, (k) FcaNet, (l) CLCFormer, (m) FsaNet, (n) SSCNet, (o) SSCBNet, and (p) FDNet (ours).

DANet improves upon these results by incorporating dual attention mechanisms, achieving an AF of 85.69 and an mIoU of 77.13%. The enhancements are particularly evident in the building and car categories, where DANet's F_1-scores are significantly higher. This underscores the importance of attention mechanisms in improving the network's focus on relevant features, thereby enhancing segmentation performance in more challenging categories. RAANet and MACU-Net, which integrate advanced attention mechanisms and multiscale feature extraction techniques, further improve performance. RAANet, for instance, achieves an AF of 86.54, with particularly strong results in the impervious surfaces and building categories, indicating its effectiveness in capturing both global and local contexts. MACU-Net similarly performs well across the board, with an OA of 85.90, demonstrating its robustness in segmenting diverse land cover types. Methods like SCAttNet and A2FPN continue this trend, achieving high F_1-scores across all categories, particularly in trees and cars, where their AF and OA values surpass 90. These results highlight the efficacy of combining spatial and channel attention mechanisms, which allow for more refined feature representation and better context modeling. Transformer-based models such as ST-UNet and CLCFormer also show strong performance, but they face challenges in fine-grained segmentation. ST-UNet, which combines the ST with a CNN backbone, achieves an AF of 91.63 and an mIoU of 81.60%, while CLCFormer, integrating CNNs and transformers, achieves an AF of 91.61 and an mIoU of 81.68%. Although these models excel in capturing global context, their performance on finer details, particularly for smaller objects like cars, is somewhat limited compared to FDNet. The superior performance of FDNet, especially in challenging categories such as low vegetation (89.97), trees (91.95), and cars (96.46), underscores its ability to handle both high-frequency and low-frequency information effectively, resulting in better boundary delineation and object segmentation.

FsaNet and SSCBNet, both of which incorporate frequency-domain processing, achieve notable performance gains, particularly in the higher frequency categories, such as cars and buildings. FsaNet, for example, reaches an AF of 92.06, while SSCBNet achieves an mIoU of 83.09%, demonstrating the potential of frequency-aware models in enhancing segmentation accuracy for complex and textured scenes. Most notably, the proposed FDNet outperforms all other methods across all evaluation metrics. FDNet achieves the highest F_1-scores in several critical categories, including low vegetation (89.97), trees (91.95), and cars (96.46), indicating its superior ability to capture and utilize both high-frequency and low-frequency information. This is further reflected in its AF of 94.01, OA of 92.47%, and mIoU of 84.92%, which represent the best results in each respective category. These outcomes validate the effectiveness of FDAM and the hierarchical integration strategy throughout the network, which together contribute to a more accurate and context-aware segmentation of RSIs.

FDNet [Fig. 5(p)] consistently delivers superior segmentation accuracy compared to the other models. It excels in capturing intricate details and preserving the structural integrity of complex urban features, such as buildings and roads, with a noticeable reduction in misclassification errors, particularly in areas where different classes are closely packed.

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TABLE IV
RESULTS ON THE LOVEDA DATASET. CLASSWISE F_1-SCORE, AF, OA, AND MIOU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Background	Building	Road	Water	Barren	Forest	Agriculture	AF	OA	mIoU
DeepLab V3+ [33]	52.29	54.99	57.16	77.96	16.11	48.18	67.79	53.50	52.30	47.62
DANet [37]	54.47	61.02	63.37	79.17	26.63	52.28	70.02	58.14	54.64	50.18
RAANet [99]	55.02	62.19	65.58	81.03	29.25	54.11	74.07	60.18	58.95	53.93
MACU-Net [100]	59.16	64.08	66.73	81.01	32.23	55.81	75.79	62.12	59.65	54.16
SCAttNet [75]	65.95	71.88	77.04	86.61	50.79	61.19	82.00	70.78	67.31	61.09
A2FPN [101]	65.17	73.32	75.19	88.01	48.82	59.96	79.71	70.03	66.89	61.14
LANet [73]	68.32	75.61	79.02	89.21	53.23	66.02	82.34	73.39	70.43	63.35
ST-UNet [84]	66.68	71.07	75.44	88.68	51.36	64.20	81.42	71.26	69.78	63.04
FcaNet [58]	66.25	72.20	77.39	87.00	51.02	61.47	82.37	71.10	67.46	61.23
CLCFormer [85]	67.17	74.34	77.69	87.71	52.34	64.91	80.96	72.16	70.45	62.55
FsaNet [54]	68.63	75.95	79.38	89.61	53.47	66.32	82.71	73.72	70.59	63.49
SSCNet [59]	69.77	75.25	80.72	89.78	55.84	68.94	84.11	74.92	70.97	64.95
SSCBNet [60]	69.45	80.18	**82.25**	91.86	53.16	65.87	84.48	75.32	72.44	65.92
FDNet (ours)	**72.17**	**82.05**	82.22	**93.54**	**56.06**	**70.19**	**85.94**	**77.45**	**75.93**	**68.94**

Compared with earlier methods such as DeepLab V3+ [Fig. 5(c)] and DANet [Fig. 5(d)], FDNet shows enhanced performance in delineating boundaries and handling small objects like cars. This improvement is particularly evident when examining high-density urban areas where precise class separation is crucial. While transformer-based methods like ST-UNet [Fig. 5(j)] and CLCFormer [Fig. 5(l)] demonstrate strong performance in capturing global context, they still face challenges in fine-grained segmentation, particularly for smaller objects. ST-UNet benefits from the ST backbone, which captures long-range dependencies, but it struggles with precise boundary delineation in areas with fine details, such as between cars and other urban elements. CLCFormer, which integrates CNNs with transformers, also exhibits strong global context understanding, but like ST-UNet, it occasionally fails to achieve the same level of precision in fine object segmentation as FDNet. In contrast, FDNet's frequency decoupling attention mechanism ensures more consistent and accurate segmentation across various urban elements, particularly in handling smaller objects and complex boundaries. This feature significantly enhances FDNet's robustness and effectiveness in real-world remote sensing applications, where precise delineation of urban structures is critical.

3) Results on the LoveDA Dataset: Table IV presents the comparative results of various state-of-the-art models on the LoveDA dataset. FDNet, as proposed in this article, demonstrates superior performance across multiple key metrics. FDNet outperforms the competing models by a significant margin in most categories, particularly excelling in the challenging "background," "building," "water," and "agriculture" classes, with F_1-score of 72.17, 82.05, 93.54, and 85.94, respectively. These results highlight FDNet's strong capability in accurately segmenting diverse land cover types, including complex structures and fine-grained regions.

FDNet achieves the highest AF of 77.45 and an OA of 75.93%, which are both substantial improvements over existing models. For instance, SSCBNet, the next best-performing model, records an AF of 75.32 and OA of 72.44%, indicating that FDNet improves segmentation accuracy across a wide range of land cover types. In addition, LANet achieve lower AF values of 73.39, further demonstrating the effectiveness of FDNet's architecture in addressing segmentation challenges in the LoveDA dataset. In comparison with transformer-based methods like ST-UNet (71.26 AF, 69.78% mIoU) and CLCFormer (72.16 AF, 70.45% mIoU), FDNet significantly outperforms in terms of both AF (94.01) and mIoU (84.92%), demonstrating its better ability to capture fine-grained details and preserve object boundaries.

In terms of mIoU, FDNet attains a value of 68.94%, outperforming other models such as SSCBNet (65.92%) and SSCNet (64.95%). This result indicates that FDNet provides a more precise prediction of segmentation boundaries, particularly in categories with complex textures and varying scales. The superior performance of FDNet is attributed to its novel frequency decoupling strategy, which effectively refines high-frequency and low-frequency features separately, leading to enhanced edge detection and contextual understanding.

Based on the visualization results presented in Fig. 6 for the LoveDA test set, FDNet [Fig. 6(o)] demonstrates superior segmentation capabilities, particularly in complex and heterogeneous environments. Compared with models like DeepLab V3+ [Fig. 6(c)] and DANet [Fig. 6(d)], FDNet excels in accurately delineating boundaries between different land cover types, such as roads, water bodies, and agricultural fields, with less confusion between similar classes. FDNet's performance is especially notable in areas with mixed or subtle transitions, such as the interfaces between barren land and forest or between roads and buildings. While models like SCAttNet [Fig. 6(g)] and A2FPN [Fig. 6(h)] show competitive results, FDNet's advanced attention mechanisms, including frequency decoupling, allow it to better capture fine details and maintain class consistency across challenging scenes, making it more effective for large-scale, high-resolution segmentation tasks in diverse landscapes.

Based on the visualization results presented in Fig. 6 for the LoveDA test set, FDNet [Fig. 6(p)] demonstrates superior segmentation capabilities, particularly in complex and heterogeneous environments. Compared with models such as DeepLab V3+ [Fig. 6(c)] and DANet [Fig. 6(d)], FDNet excels in accurately delineating boundaries between different land cover types, such as roads, water bodies, and agricultural fields, with less confusion between similar classes. The clear distinction between land types in FDNet highlights its ability

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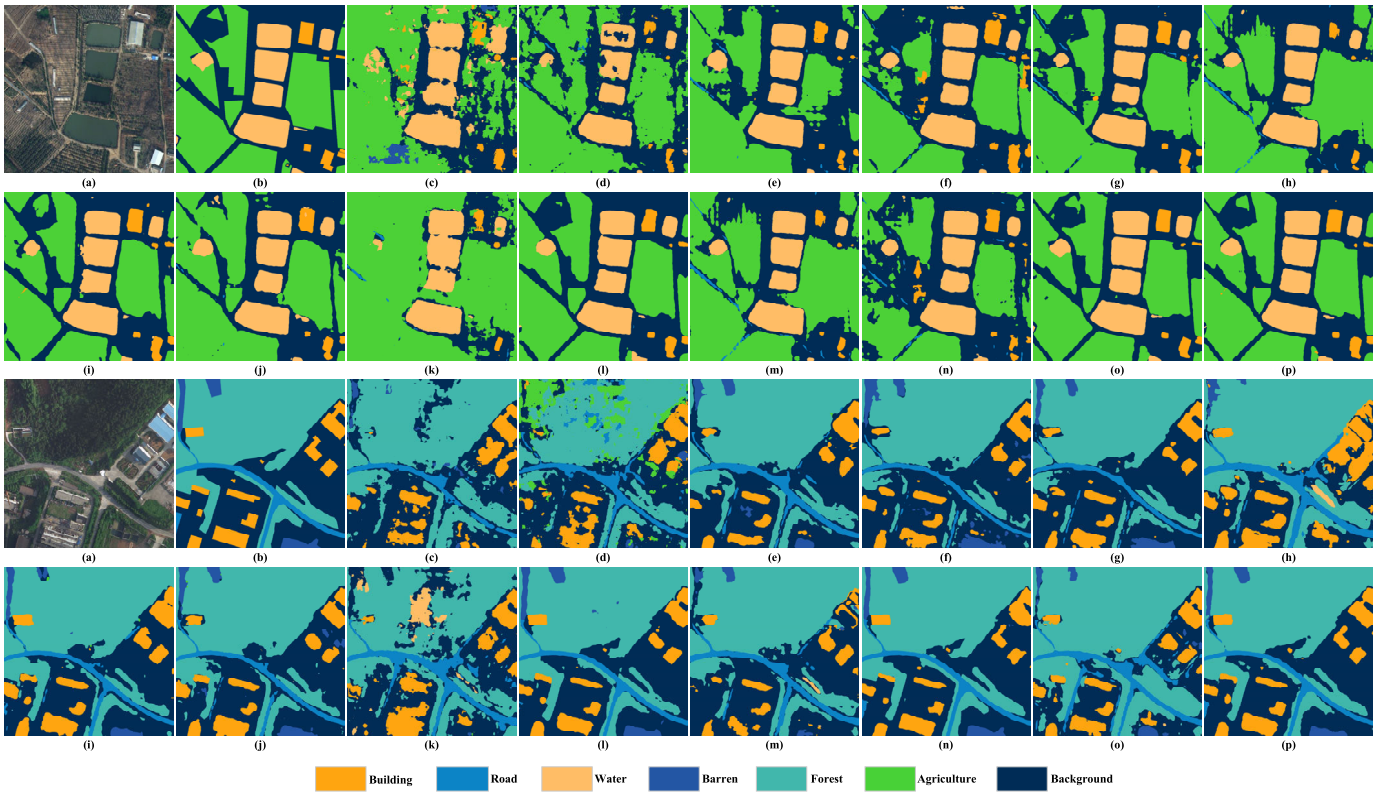


Fig. 6. Visualization comparisons on the LoveDA test set. (a) Input, (b) ground truth, (c) DeepLab V3+, (d) DANet, (e) RAANet, (f) MACU-Net, (g) SCAttNet, (h) A2FPN, (i) LANet, (j) ST-UNet, (k) FcaNet, (l) CLCFormer, (m) FsaNet, (n) SSCNet, (o) SSCBNet, and (p) FDNet (ours).

TABLE V

RESULTS ON THE DeepGlobe DATASET. CLASSWISE F_1-SCORE, AF, OA, AND mIoU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Urban land	Agriculture land	Rangeland	Forest land	Water	Barren land	Unknown	AF	OA	mIoU
DeepLab V3+ [33]	77.66	86.66	78.05	74.90	87.63	65.66	61.45	76.00	74.22	59.32
DANet [37]	79.92	87.47	79.46	76.21	88.62	67.39	63.52	77.51	75.69	62.58
RAANet [99]	79.96	88.24	79.63	75.98	88.07	67.52	62.37	77.40	75.58	63.48
MACU-Net [100]	80.37	88.01	80.42	77.04	88.55	67.32	63.95	77.95	76.12	64.95
SCAttNet [75]	76.97	87.78	77.60	80.89	89.75	70.69	68.91	78.94	77.09	65.77
A2FPN [101]	81.18	88.84	81.55	78.35	89.83	68.46	65.94	79.16	77.31	65.96
LANet [73]	79.47	90.63	80.11	80.37	87.70	71.45	71.28	80.14	78.26	66.78
ST-UNet [84]	77.84	88.77	78.47	81.80	90.75	71.48	69.69	79.83	76.95	66.51
FcaNet [58]	75.77	86.41	76.38	79.63	88.34	69.58	67.83	77.70	75.88	64.74
CLCFormer [85]	76.61	87.08	77.91	81.98	89.95	72.32	70.03	79.41	77.55	66.17
FsaNet [54]	80.87	90.01	82.01	80.55	90.66	72.58	71.06	81.11	79.20	67.57
SSCNet [59]	81.19	89.95	81.28	81.67	91.20	73.93	71.99	81.60	79.69	67.99
SSCBNet [60]	83.13	90.90	81.77	80.99	90.59	74.66	72.52	82.08	80.15	68.39
FDNet (ours)	**84.51**	**91.22**	**82.95**	**81.88**	**92.38**	**76.02**	**73.77**	**83.25**	**81.29**	**70.59**

to manage complex transitions, which is especially important in remote sensing applications. FDNet's performance is particularly notable in areas with mixed or subtle transitions, such as the interfaces between barren land and forest or between roads and buildings These areas often present challenges for other methods, but FDNet* handles them with greater precision. While models like SCAttNet [Fig. 6(g)] and A2FPN [Fig. 6(h)] show competitive results, FDNet's, allow it to better capture fine details and preserve class consistency across challenging scenes. In comparison to transformer-based methods like ST-U-Net [Fig. 6(j)] and CLCFormer [Fig. 6(l)], which capture global context well, FDNet outperforms in terms of boundary precision and segmentation consistency. While the transformer-based models demonstrate good performance in capturing long-range dependencies and global context, they sometimes struggle with fine-grained segmentation and accurate boundary delineation, particularly for smaller or adjacent land types. FDNet's superior segmentation of complex interfaces and finer details makes it more effective for large-scale, high-resolution segmentation tasks in diverse landscapes, where accuracy in class separation and boundary definition is crucial.

4) Results on the DeepGlobe Dataset: Table V presents the performance of various methods on the DeepGlobe dataset. Starting with DeepLab V3+, the baseline method demonstrates solid performance across several categories, particularly

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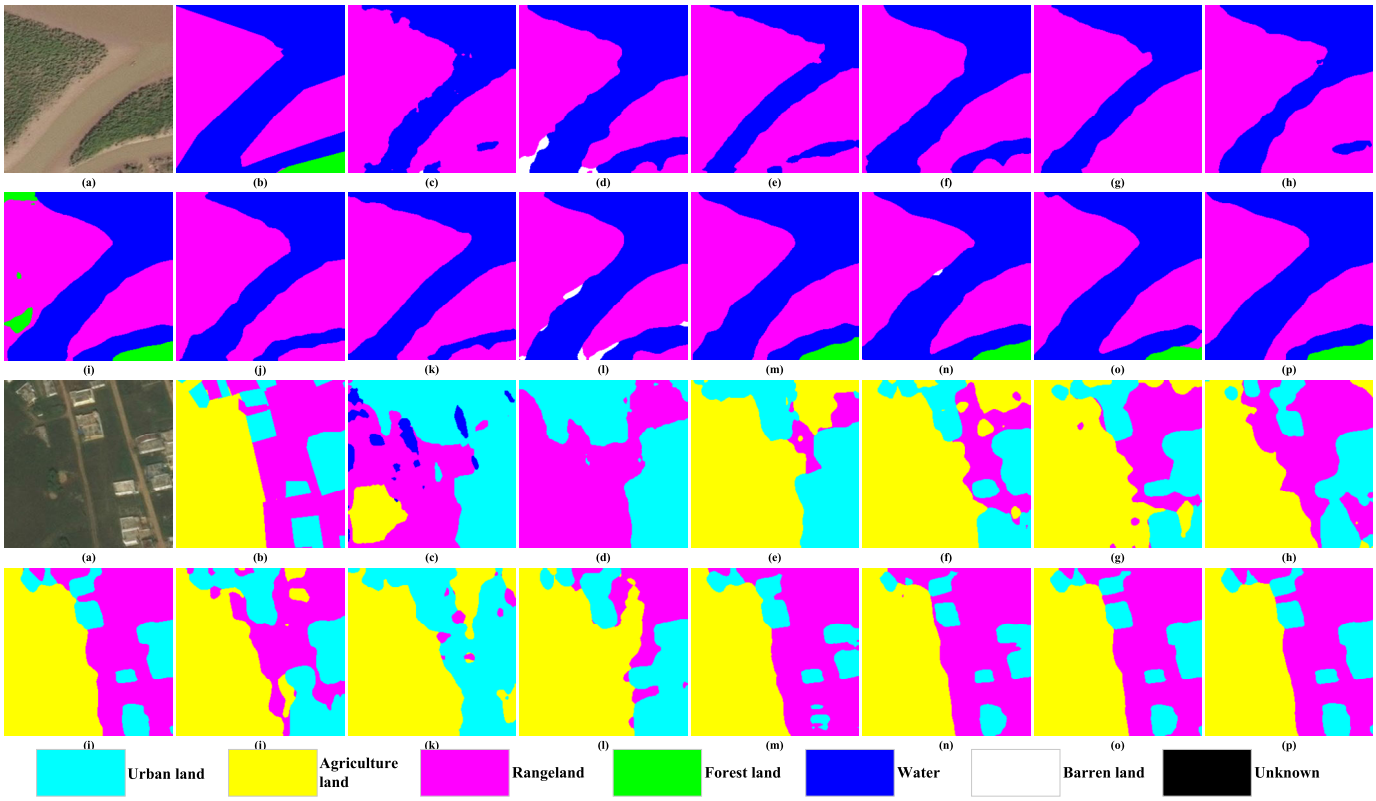


Fig. 7. Visualization comparisons on the DeepGlobe test set. (a) Input, (b) ground truth, (c) DeepLab V3+, (d) DANet, (e) RAANet, (f) MACU-Net, (g) SCAttNet, (h) A2FPN, (i) LANet, (j) ST-UNet, (k) FcaNet, (l) CLCFormer, (m) FsaNet, (n) SSCNet, (o) SSCBNet, and (p) FDNet (ours).

in agriculture land and water, where it achieves F_1-scores of 86.66 and 87.63, respectively. However, its performance is less consistent in other categories such as urban land and barren land, leading to an overall AF of 76.00 and mIoU of 59.32%. These results suggest that while DeepLab V3+ is effective for general segmentation tasks, it struggles to generalize across the diverse land cover types present in the DeepGlobe dataset.

DANet, which incorporates a dual attention mechanism, shows a noticeable improvement over DeepLab V3+, achieving higher F_1-scores in urban land, agriculture land, and water. Its AF improves to 77.51 and mIoU to 62.58%, indicating that the attention mechanisms help the model to better capture relevant features in complex scenes, thus improving segmentation performance. RAANet and MACU-Net further build on these improvements, with RAANet achieving an AF of 77.40 and MACU-Net an AF of 77.95. Both methods show strong performance across a range of categories, particularly in agriculture land and water, demonstrating their robustness in handling high-resolution satellite imagery. MACU-Net's performance is particularly notable in the forest land category, where it achieves an F_1-score of 77.04, suggesting its effectiveness in capturing intricate details in natural environments.SCAttNet and A2FPN demonstrate even more competitive results, particularly in forest land and barren land categories. SCAttNet achieves an AF of 78.94 and mIoU of 65.77%, while A2FPN slightly edges ahead with an AF of 79.16 and mIoU of 65.96%. These results underscore the benefits of integrating spatial and channel attention mechanisms, which allow these models to more effectively distinguish between different land cover types and improve overall segmentation accuracy. In comparison with transformer-based methods such as ST-UNet (AF: 79.83 and mIoU: 66.51) and CLCFormer (AF: 79.41 and mIoU: 66.17), FDNet significantly outperforms in key metrics, achieving an AF of 83.25 and mIoU of 70.59. The superior performance of FDNet highlights its ability to more effectively capture both local and global contextual information, especially in complex land cover classes, making it particularly suitable for high-resolution segmentation tasks.

FsaNet and SSCNet, which leverage frequency domain processing, achieve significant gains in performance. FsaNet attains an AF of 81.11 and mIoU of 67.57%, indicating its strength in managing complex textures and subtle spectral variations. SSCNet performs slightly better, with an AF of 81.60 and mIoU of 67.99%, particularly excelling in categories like forest land and water, where the frequency-based approach proves highly effective in enhancing feature representation and segmentation accuracy. The proposed FDNet surpasses all other methods, delivering the highest performance across all metrics. FDNet achieves the best F_1-scores in multiple categories, including urban land (84.51), agriculture land (91.22), rangeland (82.95), forest land (81.88), water (92.38), and barren land (76.02). These results contribute to an AF of 83.25, an OA of 81.29%, and an mIoU of 70.59%, establishing FDNet as the most effective model on the DeepGlobe dataset. The exceptional performance of FDNet can be attributed to its innovative FDAM, which effectively captures and refines both high-frequency and low-frequency components, thereby enhancing the network's ability to accurately segment complex and diverse land cover types.

As shown in Fig. 7, FDNet [Fig. 7(p)] shows a marked improvement in segmenting diverse land cover types, particularly in challenging categories like rangeland, barren

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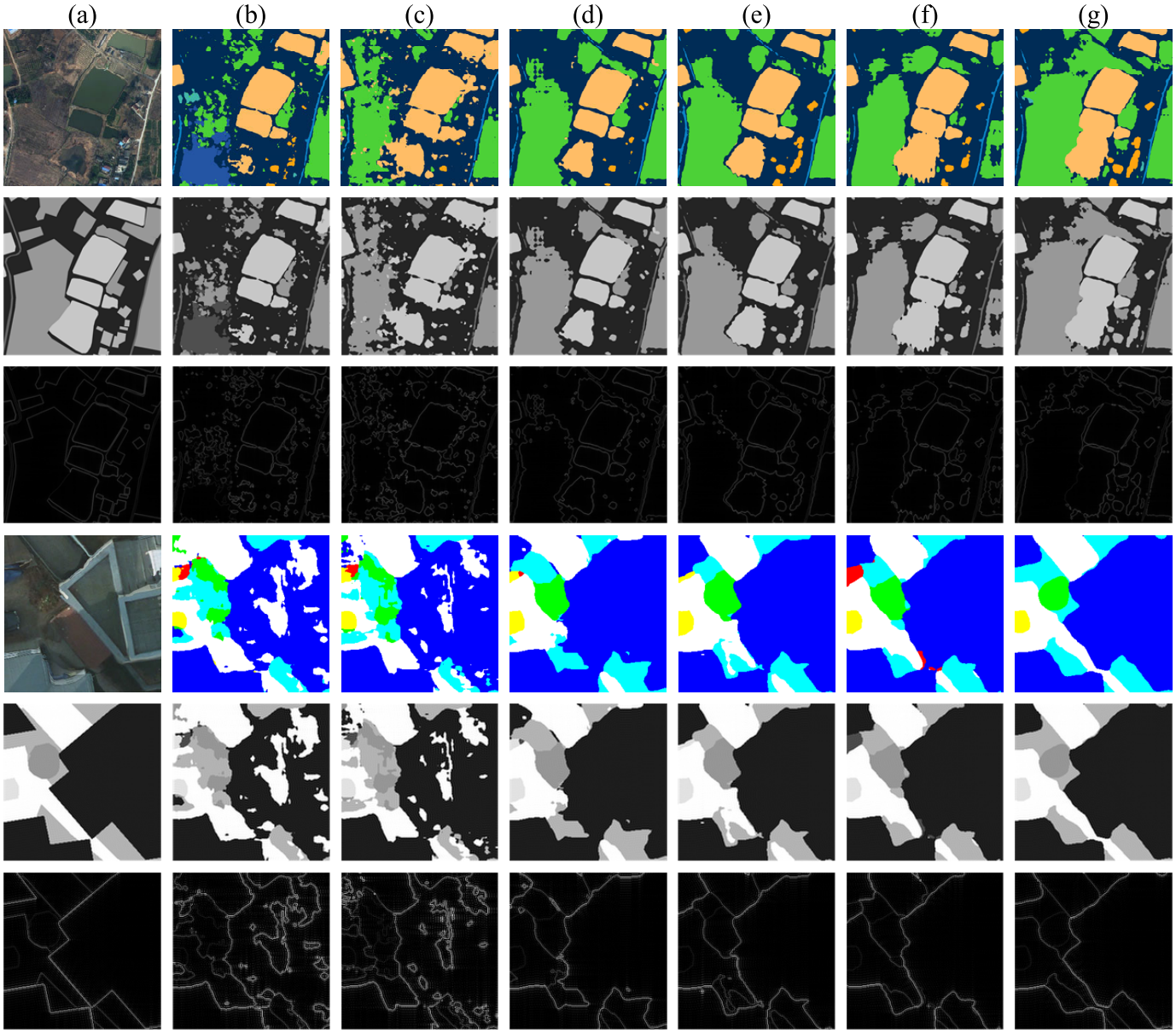


Fig. 8. Visualization comparisons of ablation study. (a) Input, (b) FCN + SE [35], (c) FCN + CBAM [36], (d) FCN + DAM [37], (e) FCN + LAM [72], (f) FCN + SAPM [77], and (g) FCN + AFCM (ours). The second and third rows compare the results of low-frequency and high-frequency distributions.

land, and mixed-use areas. Compared to earlier methods such as DeepLab V3+ [Fig. 8(c)] and DANet [Fig. 8(d)], FDNet delivers more accurate boundary delineation and reduces class confusion, especially in areas where different land types closely intersect. The advanced attention mechanisms in FDNet, particularly its frequency decoupling approach, enable it to better capture fine details and subtle transitions between land cover classes. This results in more consistent and reliable segmentation outputs, even in regions with complex spatial structures or overlapping features.

While other models such as SCAttNet [Fig. 8(g)] and A2FPN [Fig. 8(h)] perform well, FDNet consistently provides clearer and more accurate segmentation, underscoring its effectiveness for large-scale land cover classification tasks in diverse and intricate environments. In comparison to transformer-based methods such as ST-UNet [Fig. 8(j)] and CLCFormer [Fig. 8(l)], FDNet shows superior precision, particularly in boundary refinement and fine-grained segmentation, where the transformer-based models sometimes struggle with subtle class transitions or fine details.

E. Ablation Experiments of AFCM

1) Numerical Comparisons: Table VI presents the results of embedding different attention models at the end of the encoder within an FCN-based framework across four benchmark datasets. In our experiments, the hyperparameter settings are carefully chosen to ensure optimal performance while maintaining consistency with widely adopted practices in the field. Specifically, we adopt the encoder–decoder depth and convolutional kernel size from the SegNet [30] framework, which is based on the ResNet101 [95] backbone, as it provides a robust balance between computational efficiency and

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TABLE VI
RESULTS OF EMBEDDING DIFFERENT ATTENTION MODELS. NUMERICAL COMPARISONS ARE PRESENTED IN THE FORM OF AF/OA/mIoU, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Models	Vaihingen	Potsdam	LoveDA	DeepGlobe
FCN [29]	68.30/66.67/62.38	69.58/68.81/64.47	50.32/49.75/46.54	68.59/66.98/53.54
FCN+SE [35]	69.15/67.49/63.20	70.25/69.54/65.13	51.10/50.60/47.30	69.30/67.85/54.15
FCN+CBAM [36]	69.96/68.07/63.77	70.60/69.90/65.67	52.23/51.11/48.05	69.97/68.44/54.82
FCN+DAM [37]	69.90/68.40/64.30	70.65/69.55/65.20	52.20/51.60/48.50	70.32/68.54/55.10
FCN+LAM [73]	71.39/69.79/65.53	72.97/71.23/66.61	53.33/52.77/49.52	71.83/70.91/56.22
FCN+SAPM [78]	72.05/70.41/66.79	74.15/72.85/67.97	54.61/54.02/50.91	73.15/72.35/57.53
FCN+AFCM (ours)	**73.47/71.89/68.00**	**75.60/74.37/69.23**	**56.28/55.78/52.62**	**74.59/73.95/59.58**

segmentation accuracy. The weighting coefficients for high- and low-frequency components in the frequency attention (FA) module, as well as the attention fusion weighting factor, are set to 0.5.

Starting with the baseline FCN model, it is evident that while the model performs reasonably well across all datasets, it exhibits limitations in capturing complex spatial relationships and contextual dependencies, as reflected in the relatively low AF, OA, and mIoU. For instance, the baseline FCN achieves an AF of 68.30, an OA of 66.67%, and an mIoU of 62.38% on the Vaihingen dataset, indicating a need for more sophisticated feature refinement mechanisms.

When SE block is integrated into the FCN, a modest improvement is observed across all datasets. The SE block enhances the network's ability to recalibrate channel-wise feature responses, leading to better performance, particularly in the DeepGlobe dataset, where the AF improves to 69.30 and the mIoU to 54.15%. However, the overall gains remain incremental, suggesting that while SE blocks provide some benefit, they may not be sufficient to capture the full complexity of the datasets.

The addition of CBAM and DAM further improves performance, particularly in more challenging datasets, such as LoveDA and DeepGlobe. These models leverage both spatial and channel attention, allowing the network to focus on more relevant features while suppressing less important ones. For example, FCN + CBAM achieves an AF of 69.96 on the Vaihingen dataset, with a corresponding mIoU of 63.77%, indicating a more refined feature extraction process. Similarly, FCN + DAM achieves slightly better results across most datasets, reflecting the advantages of combining spatial and channel attention mechanisms.

The integration of LAM and SAPM leads to more substantial improvements. LAM enhances the model's ability to capture local spatial dependencies, while SAPM provides a multiscale approach to spatial attention, which is particularly effective in diverse and complex scenes. On the Potsdam dataset, for instance, FCN + SAPM achieves an AF of 74.15, an OA of 72.85%, and an mIoU of 67.97%, outperforming the previous models and highlighting the importance of multiscale spatial attention in RSI segmentation.

The proposed AFCM, when integrated into the FCN, yields the best performance across all evaluated datasets. The AFCM model achieves superior results with an AF of 73.47, an OA of 71.89%, and an mIoU of 68.00% on the Vaihingen dataset, and consistently outperforms other models on the Potsdam, LoveDA, and DeepGlobe datasets as well. These improvements can be attributed to AFCM's ability to decouple frequency components and apply refined attention mechanisms that capture both global and local contextual information across multiple scales. The significant performance gains, particularly in complex datasets like LoveDA (AF: 56.28 and mIoU: 52.62%) and DeepGlobe (AF: 74.59 and mIoU: 59.58%), demonstrate AFCM's effectiveness in enhancing feature representation and segmentation accuracy.

2) Visual Comparisons of Ablation Study of AFCM: To demonstrate the impact of the AFCM on feature refinement in the frequency domain, we conducted a visual comparison between different attention mechanisms integrated into the FCN architecture. These include SE [35], CBAM [36], DAM [37], LAM [72], and SAPM [77]. The visualizations highlight the contributions of these methods in refining low-frequency and high-frequency distributions for feature extraction. Fig. 8 illustrates the significant improvement achieved by AFCM in modeling both frequency components. The low-frequency visualizations demonstrate how AFCM effectively captures global structures, while the high-frequency representations reveal enhanced edge preservation and fine-grained texture extraction. Compared to alternative methods, AFCM achieves a balanced refinement of both frequency domains, ensuring superior contextual modeling and structural consistency.

F. Ablation Experiments of SSEM and FDAM

To analyze the contributions of the SSEM and the FDAM within FDNet, we conducted ablation experiments across four benchmark datasets: ISPRS Vaihingen, ISPRS Potsdam, LoveDA, and DeepGlobe. The results of these experiments, shown in Table VII, are presented in terms of AF, OA, and mIoU.

The first experiment evaluates the effect of using SSEM without FDAM. In this configuration, FDNet achieves moderate performance, with results such as AF/OA/mIoU of 86.54/83.81/79.61 on the Vaihingen dataset and 72.05/70.34/62.36 on LoveDA. These scores indicate that while SSEM contributes to improving feature learning by focusing on the most informative spectral bands, it alone is insufficient for optimal segmentation, particularly in complex datasets such as LoveDA and DeepGlobe.

The second experiment isolates the impact of FDAM without SSEM. In this setup, FDNet exhibits significant improvements across all datasets, achieving an AF/OA/mIoU of 89.77/88.13/82.59 on Vaihingen and 74.75/72.99/64.68 on

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TABLE VII
RESULTS OF ABLATION EXPERIMENTS OF SSEM AND FDAM. NUMERICAL COMPARISONS ARE PRESENTED IN THE FORM OF AF/OA/MIOU

Method	SSEM	FDAM	Vaihingen	Potsdam	LoveDA	DeepGlobe
FDNet	✓		86.54/83.81/79.61	89.20/87.77/80.89	72.05/70.34/62.36	78.66/76.78/66.83
FDNet		✓	89.77/88.13/82.59	92.55/90.93/83.83	74.75/72.99/64.68	81.58/79.60/69.22
FDNet	✓	✓	**92.05/91.05/84.09**	**94.01/92.47/84.92**	**77.45/75.93/68.94**	**83.25/81.29/70.59**

TABLE VIII
COMPARISON OF MODEL SIZE, FLOPS, AND FPS FOR DIFFERENT ATTENTION MODELS

Models	Parameters (Mb)	FLOPs (Gbps)	FPS (Frames/s)
FCN [29]	23.79	21.95	84.7
FCN+SE [35]	23.80	21.95	83.9
FCN+CBAM [36]	23.97	21.95	80.3
FCN+DAM [37]	25.75	28.01	71.4
FCN+LAM [73]	23.80	21.98	82.5
FCN+SAPM [78]	26.65	22.36	78.8
FCN+AFCM (ours)	27.80	23.55	75.2

LoveDA. FDAM's ability to refine high-frequency and low-frequency components separately improves edge detection and segmentation accuracy, especially in challenging RSIs with diverse textures and intricate boundaries.

When both SSEM and FDAM are used together, FDNet achieves its best performance, demonstrating the complementary effects of the two modules. The full model achieves the highest AF/OA/mIoU scores of 92.05/91.05/84.09 on Vaihingen, 94.01/92.47/84.92 on Potsdam, 77.45/75.93/68.94 on LoveDA, and 83.25/81.29/70.59 on DeepGlobe. These results clearly show that the integration of SSEM and FDAM significantly enhances FDNet's ability to capture both spectral and spatial details, leading to more accurate segmentation across a variety of datasets.

The ablation experiments validate the necessity of both SSEM and FDAM for achieving state-of-the-art performance. SSEM improves spectral feature learning by compressing redundant information, while FDAM refines spatial features by focusing on different frequency components. Together, these modules allow FDNet to outperform alternative approaches, particularly in complex environments with diverse land cover types.

G. Efficiency Analysis of AFCM

Table VIII lists the comparison of the model size, floating point operations (FLOPs), and frames per second (FPS) across different attention mechanisms integrated into the FCN architecture on the ISPRS Vaihingen dataset. The FLOPs calculations reflect the computational cost of processing 256 × 256 image patches with a batch size of 32. The base FCN model, with 23.79 Mb of parameters, 21.95 Gb/s of FLOPs, and an FPS of 84.7, serves as the baseline. Simpler attention mechanisms, such as SE and CBAM, introduce minimal computational overhead, maintaining similar parameters (23.80 and 23.97 Mb, respectively) and FLOPs (both at 21.95 Gb/s), while achieving FPS values of 83.9 and 80.3, respectively. This demonstrates their efficiency but also highlights limited enhancement in feature refinement.

More sophisticated modules like DAM and SAPM significantly increase computational requirements due to their complex attention operations. DAM raises the parameter count to 25.75 Mb and FLOPs to 28.01 Gb/s, with a reduced FPS of 71.4, indicating higher processing demands. Similarly, SAPM reaches 26.65 Mb in parameters and 22.36 Gbps in FLOPs, with an FPS of 78.8, balancing improved feature representation at the cost of processing speed. These results underscore the tradeoffs inherent in employing more advanced attention mechanisms.

The proposed AFCM strikes a balance between complexity and performance. While introducing a modest increase in parameters (27.80 Mb) and FLOPs (23.55 Gb/s) compared with simpler modules, AFCM maintains an acceptable FPS of 75.2. This efficiency is attributed to its innovative frequency decoupling and multiscale attention design, which enhance feature extraction without proportionately increasing computational demands. As a result, AFCM offers a robust and efficient solution for high-resolution semantic segmentation tasks, making it practical for real-world remote sensing applications where both accuracy and efficiency are critical.

V. CONCLUSION

This article presents FDNet, a pioneering FDNet that significantly advances the semantic segmentation of RSIs by effectively leveraging the distinct characteristics of high- and low-frequency components. Our approach addresses the limitations of traditional spatial-centric models by incorporating advanced spectral feature optimization techniques that effectively capture and refine the intricate details present in complex RSI data. Through the integration of the SSEM, the FDAM, and the AFCM, FDNet achieves significant improvements in segmentation accuracy and generalization across diverse datasets.

Our extensive experimental results demonstrate that FDNet consistently outperforms existing state-of-the-art methods, particularly in challenging environments characterized by diverse land cover types and high-resolution imagery. The frequency decoupling strategy not only improves edge preservation and detail recognition but also enhances the model's ability to distinguish between similar land cover classes with subtle spectral differences. Moreover, the hierarchical embedding of AFCMs within FDNet contributes to its robustness and adaptability, making it a powerful tool for large-scale geospatial analysis.

Future work will focus on further optimizing the computational efficiency of FDNet, exploring its application to real-time processing scenarios, and extending its capabilities to other remote sensing tasks, such as object detection and change detection. In addition, we will investigate the potential of integrating FDNet with emerging neural network architectures to further enhance its performance and scalability.

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